

16. If n is a root of the equation $x^2 + px + m = 0$ and m is a root of the equation $x^2 + px + n = 0$, where $m \neq n$, then what is the value of $p + m + n$?

- (a) -1
- (b) 0
- (c) 1
- (d) 2

17. In how many ways can a student choose $(n-2)$ courses out of n courses if 2 courses are compulsory ($n > 4$)?

- (a) $(n-3)(n-4)$
- (b) $(n-1)(n-2)$
- (c) $(n-3)(n-4)/2$
- (d) $(n-2)(n-3)/2$

18. If

$$D_n = \begin{vmatrix} n & 20 & 30 \\ n^2 & 40 & 50 \\ n^3 & 60 & 70 \end{vmatrix}$$

then what is the value of $\sum_{n=1}^4 D_n$?

- (a) -10000
- (b) -10
- (c) 10
- (d) 10000

19. Consider the following in respect of the matrices

$$P = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \text{ and } Q = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

- I. PQ is a null matrix.
- II. QP is an identity matrix of order 3.
- III. $PQ = QP$

Which of the above is/are correct?

- (a) I only
- (b) II only
- (c) I and III
- (d) II and III

20. If P is a skew-symmetric matrix of order 3, then what is $\det(P)$ equal to?

- (a) -1
- (b) 0
- (c) 1
- (d) 3

